MedGraphRAG: A Safety-First Hybrid Graph Retrieval-Augmented Generation System for Offline Clinical Decision Support

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Abstract

Background and Objective. Large language models (LLMs) deployed in clinical decision support face a safety-utility trade-off: unconstrained answering yields unacceptable rates of wrong assertions, whereas excessive refusal renders systems inert. Existing retrieval-augmented generation (RAG) frameworks lack mechanistic, bounded safety controls deployable on institutional hardware. We present MedGraphRAG, an offline hybrid GraphRAG system that reframes clinical AI evaluation from unconstrained accuracy maximisation to bounded wrong-assertion rate (WAR).

Methods. MedGraphRAG couples dense vector retrieval (FAISS IVF-PQ with MedCPT embeddings) and lexical retrieval (BM25) with an embedded property graph (Kuzu) enriched with 632,931 relational edges (617,219 NEGATES and 15,712 TEMPORAL_BEFORE). Dual candidate streams are merged via Reciprocal Rank Fusion and re-ranked with an additive relation-aware score ($\gamma = 0.15$). A beta-gated verification layer enforces a strict WAR $\leq 10\%$ constraint by combining evidence faithfulness (φ) and graph consistency (S_g): $V = \beta\varphi + (1-\beta)S_g$. The system executes entirely offline on a commodity laptop (Ryzen 7 7840HS / RTX 3050 6 GB) using Qwen2.5-7B-Instruct Q4_K_M. Private health data are encrypted with AES-256-GCM and HKDF-derived keys. Retrospective evaluation was performed on MedQA-US ($N = 500$, seed 42, $T = 0.0$).

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Results. At the calibrated safe threshold ($\tau^* = 0.80$), Graph-Only mode (M2) achieves 60.8% precision-on-answered with WAR = 8.0% (79.6% refusal). Disabling the gate drives WAR to 44.6% across all 500 queries. Deterministic query rewriting expands answered coverage by 3.2–3.4 pp while maintaining bounded error. Hybrid retrieval completes in 455.7 ms (95% CI: [388.4, 495.8] ms); full end-to-end pipeline latency is 4,083 ms (M2) to 12,667 ms (M4).

Conclusion. MedGraphRAG demonstrates that bounded clinical error and interpretable offline decision support are jointly achievable on consumer hardware without cloud dependencies or GPU clusters.

Keywords: GraphRAG, clinical-NLP, hallucination-mitigation, knowledge-graph, retrieval-augmented-generation, decision-support, safety

1. Introduction

Deploying large language models (LLMs) as point-of-care clinical assistants offers direct utility for diagnostic and therapeutic workflows (Singhal et al., 2023; Thirunavukarasu et al., 2023). However, the tendency of ungrounded parametric architectures to hallucinate plausible yet factually incorrect assertions poses direct patient-safety hazards (Bai et al., 2022). In high-stakes medicine, confident assertions regarding contraindicated medications, misidentified diagnostic criteria, or inverted pathological chronologies can cause serious clinical harm.

To mitigate hallucinations, retrieval-augmented generation (RAG) grounds language generation on passages retrieved from external corpora (Lewis et al., 2020; Gu et al., 2021). Standard medical RAG frameworks rely on dense bi-encoders (Jin et al., 2023; Gao et al., 2023), which match semantic similarity effectively but struggle with complex clinical syntax.

Specifically, vector embeddings frequently compress or lose negation scope (e.g., conflating "no evidence of pulmonary embolism" with "pulmonary embolism") and fail to enforce multi-step temporal chronologies required in differential diagnosis (Han et al., 2022; Mallen et al., 2023).

Knowledge graphs provide a complementary symbolic representation, structuring clinical entities and ontological dependencies into explicit triples (Bodenreider, 2004; Yasunaga et al., 2021). Recent hybrid GraphRAG systems combine graph structures with textual retrieval (Edge et al., 2024; He et al., 2024; Wu et al., 2024) (Wu et al. share the MedGraphRAG name but focus on entity-centric graph construction; our architecture emphasizes hybrid retrieval fusion and dual-gated verification). Nevertheless, existing medical GraphRAG implementations exhibit three fundamental limitations:

First, current evaluations focus almost exclusively on unconstrained accuracy (e.g., aggregate multiple-choice scores), omitting explicit bounds on incorrect statements. In clinical practice, an assistant that answers 60% of questions correctly while making wrong assertions on 40% is clinically unacceptable. Safe deployment requires a constrained-optimisation framework that caps the wrong-assertion rate (WAR) below an acceptable tolerance ϵ, treating refusal (abstention) as an active safety mechanism rather than system failure.

Second, existing biomedical knowledge graphs capture predominantly positive ontological relations (e.g., TREATS or CAUSES), neglecting negation and temporal progression. Without these primitives, retrieval graphs cannot detect when retrieved passages contain superseded diagnoses or explicit exclusions.

Third, most GraphRAG systems rely on cloud-hosted language models (e.g., GPT-4) and server clusters. This architecture violates patient data privacy regulations (e.g., HIPAA, GDPR) that prohibit transmitting protected health information (PHI) beyond institutional boundaries. Practical point-of-care deployment requires secure, sub-second hybrid retrieval and inference executing locally on commodity hardware.

We present **MedGraphRAG**, an offline hybrid GraphRAG system designed for point-of-care clinical decision support. Our primary contributions are:

1. **Safety-Constrained Pareto Calibration:** A verification framework optimising answered accuracy subject to WAR $\leq$ 10%. Across 15 threshold settings on MedQA-US ($N = 500$), we establish the Pareto frontier, demonstrating that ungated generation yields unacceptable error rates ($\approx$ 44%).
2. **Relational Graph Enrichment:** Ingestion of 632,931 negation and temporal relational edges into an embedded Kuzu graph, demonstrating mechanistically how symbolic consistency checks overturn hallucinations bypassing dense retrieval.
3. **Deterministic Query Rewriting and Relation Reranking:** A temperature-zero query normaliser improving answered coverage by 3.2–3.4 pp at matched thresholds, combined

with a relation-aware reranker boosting Graph-Hit@5 by 13.6 pp.

4. **Hardened Offline Stack on Consumer Hardware:** A self-contained system operating on an NVIDIA RTX 3050 (6 GB) laptop GPU, achieving hybrid retrieval in 455.7 ms with authenticated AES-256-GCM encryption and a resolved 13-point security audit.

Supplementary materials provide the DECIDE-AI applicability mapping, IJMI ML checklist, and extended tables; configuration details appear in Appendix A.

2. Methods

2.1. System Architecture

Figure 1 shows the MedGraphRAG inference pipeline. Stage 1 deterministically standardises input queries into clinical phrasing. Stage 2 executes dual-stream dense and lexical retrieval, combining candidates via Reciprocal Rank Fusion and relation-aware graph reranking. Stage 3 expands clinical entities within the embedded Kuzu graph. Stage 4 computes composite beta-gated verification scores, triggering structured refusals for low-confidence queries. Stage 5 generates evidence-grounded responses using a quantized LLM isolated behind a GPU mutex lock.

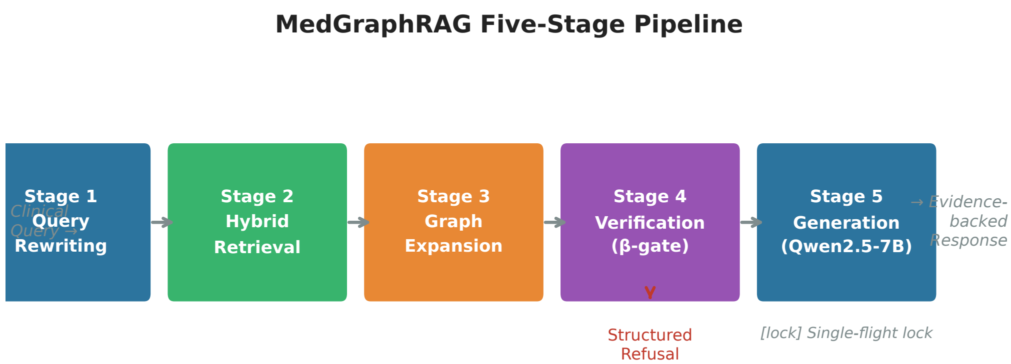

Figure 1: MedGraphRAG five-stage offline pipeline architecture. Stage 1 executes deterministic query rewriting ($T = 0.0, \leq 30$ words). Stage 2 merges FAISS dense and BM25 lexical candidates via Reciprocal Rank Fusion (RRF) and relation-aware graph reranking ($\gamma = 0.15$). Stage 3 traverses the Kuzu property graph. Stage 4 applies beta-gated verification ($\beta = 0.7$) enforcing WAR $\leq 10\%$. Stage 5 serializes LLM generation behind a hardware mutex lock.

2.2. Deterministic Query Rewriting

Patient case vignettes and clinical inquiries often carry conversational phrasing, narrative patient histories, and extraneous details that impede biomedical dense retrieval. Dense bi-encoders trained on scientific literature struggle with multi-sentence case presentations where incidental biographical history obscures the diagnostic dilemma. Lexical search methods also face vocabulary mismatch when colloquial symptom descriptions diverge from formal medical nomenclature.

To standardise heterogeneous queries without stochastic variance, Stage 1 uses Qwen2.5-7B-Instruct executing strictly at temperature $T = 0.0$. The rewriter uses a structured system prompt directing the model to extract the primary pathological condition, salient diagnostic findings, and therapeutic targets into a canonical clinical summary capped at 30 words. For example, a narrative vignette describing an elderly patient who experiences severe rotational vertigo triggered by lateral head motion is transformed into the canonical clinical query: "benign paroxysmal positional vertigo diagnostic criteria and Dix-Hallpike confirmatory maneuvers." Setting temperature $T = 0.0$ alongside greedy decoding guarantees mathematical determinism: identical clinical inputs generate bit-identical textual reformulations across successive executions, fulfilling mandatory auditability and reproducibility requirements for clinical decision support software.

2.3. Dual-Stream Hybrid Retrieval and Reranking

Retrieval operates over an extensive clinical reference corpus partitioned into coherent passages of approximately 512 tokens. The dense retrieval stream encodes passages using the MedCPT query and article bi-encoder architecture (Jin et al., 2023), which was pre-trained on hundreds of millions of PubMed search query-article click logs to optimize zero-shot biomedical retrieval. Chunk vectors are indexed and searched using FAISS IVF-PQ (Inverted File with Product Quantisation; Johnson et al., 2021), partitioning the vector space into coarse Voronoi centroids to achieve sub-linear retrieval speed. Concurrently, a parallel lexical stream evaluates candidate documents using the classical BM25 probabilistic relevance framework (Robertson and Zaragoza, 2009), operating over a dense candidate pool of 200 retrieved documents.

Because dense and lexical retrievers exhibit complementary error profiles (dense models capture latent semantic affinity, whereas lexical models enforce exact keyword and numerical matches), their respective rank distributions R_{dense} and R_{lex} are integrated via Reciprocal Rank Fusion (RRF; Cormack et al., 2009):

$$s_{\text{fused}}(d) = \sum_{m \in \{\text{dense,lex}\}} \frac{1}{k_{\text{rrf}} + r_m(d)}, \tag{1}$$

where $r_m(d)$ denotes the ordinal rank of candidate passage d within stream m, and $k_{\text{rrf}} = 60$ represents the standard monotonic smoothing constant that prevents top-ranked outliers in a single stream from dominating the fused ranking.

To bias retrieval toward passages with corroborated topological connections to the clinical entities identified in the query, we implement an additive relation-aware graph reranking function:

$$s_{\text{hybrid}}(d) = s_{\text{fused}}(d) + \gamma \cdot r_{\text{rel}}(d), \tag{2}$$

where $r_{\text{rel}}(d) = \sum_{e \in \mathcal{E}(d)} w(\text{type}(e))$ aggregates the domain-weighted edge counts connecting entities cited within candidate passage d to the query seed entities within the property graph. The relative importance of edge categories is parameterized by clinical specificity weights (e.g., pharmacological treatments and adverse effects receive weight 1.0, diagnostic lab associations and negation edges receive 0.8, temporal sequences receive 0.6, and broad taxonomic hierarchies receive 0.3; Appendix A). The global multiplier $\gamma = 0.15$

balances semantic similarity with structural graph reinforcement, calibrated empirically via controlled ablations (Section 3.3).

2.4. Relational Knowledge Graph Enrichment

Structured relations are stored in Kuzu (Feng et al., 2023), an embeddable, disk-backed graph database supporting openCypher queries. While standard medical graphs capture static disease-symptom linkages, clinical decision support requires modeling negation and temporal order.

We processed 2,294,038 text chunks using rule-guided linguistic parsing grounded in UMLS (Bodenreider, 2004). Operating at 0.2329 ms per chunk (534.45 s total wall-clock), the pipeline ingested **632,931** directed relational edges:

- **NEGATES Edges (617,219):** Linking text chunks to entities bearing explicit negation cues ("no evidence of," "ruled out," "contraindicated in"), comprising 111,720 Chunk $\rightarrow$ Disease, 21,001 Chunk $\rightarrow$ Drug, and 484,498 Chunk $\rightarrow$ OntologyTerm edges.
- **TEMPORAL_BEFORE Edges (15,712):** Linking clinical concepts via chronological sequence cues ("prior to," "following," "within one hour of"), comprising 12,225 Disease $\rightarrow$ Disease and 3,487 Drug $\rightarrow$ Drug edges.

Manual auditing of sampled endpoints confirmed high directional validity. A cryptographic SHA-256 retrieval signature verified 0.00% retrieval drift on baseline queries post-ingestion (Appendix A).

2.5. Beta-Gated Verification and Pareto Threshold Calibration

Standard RAG architectures lack explicit confidence gates to arbitrate whether retrieved evidence warrants generating an answer. Unhedged language models produce fluent assertions even when underlying evidence is tenuous, contradictory, or absent. To prevent clinical hallucinations, MedGraphRAG adds a dual-evidence verification layer that evaluates candidate response fidelity prior to releasing output.

For each candidate assertion, the verification module evaluates a composite verification metric $V \in [0, 1]$ that combines passage-level entailment with symbolic graph consistency:

$$V = \beta\,\varphi + (1 - \beta)\,S_g, \tag{3}$$

where $\varphi \in [0, 1]$ denotes an NLI-derived faithfulness score measuring the degree of semantic entailment between the generated assertion and the retrieved reference passages (operating via normalised MedCPT embedding cosine similarity as an efficient deterministic proxy when an independent cross-encoder NLI judge is omitted), and $S_g \in [0, 1]$ denotes the structural graph consistency score. The graph consistency metric S_g evaluates the ratio of asserted clinical entity triples that are corroborated by non-negated multi-hop paths within a 2-hop radius in Kuzu, while applying an immediate penalizing discount whenever an asserted entity matches a NEGATES edge. The balance hyperparameter $\beta = 0.70$ allocates primary decision weight to passage evidence while preserving strong sensitivity to symbolic graph contradictions.

When the composite score falls below a calibrated refusal threshold τ, generation is suppressed, and the system emits a structured clinical refusal detailing the precise nature of the missing evidence. We calibrate τ through a constrained optimisation problem:

$$\tau^* = \arg\max_{\tau} \ \mathrm{Acc}(\mathrm{all}) \quad \text{subject to} \quad \mathrm{WAR}(\tau) \leq 10\%, \tag{4}$$

where Acc(all) represents the proportion of correct responses across the entire evaluation cohort ($N = 500$), and WAR represents the proportion of wrong assertions across the same cohort. We sweep 15 discrete threshold candidate points ($\tau \in \{0.10, 0.15, 0.20, \ldots, 0.80\}$), identifying the Pareto-optimal operating threshold τ^* that maximises clinically useful answers while strictly guaranteeing that erroneous assertions remain within the institutional safety threshold.

2.6. Safety Guardrails

Two deterministic guardrails operate at sub-millisecond latency:

- **F1: Cross-Modal Discrepancy Guardrail.** In multimodal contexts, Rule A detects conflicts when an imaging finding score ≥ 0.60 contradicts a text negation cue for the same finding (e.g., visual pneumothorax detection of 0.88 versus report text “clear lungs”). Rule B detects positive textual assertions that contradict negated imaging observations. F1 escalates conflicts by issuing a high-severity alert recommending specialist review rather than attempting autonomous resolution.

- **F2: Epistemic Knowledge-Gap Mapper.** When a query is refused, F2 classifies the failure into three gap types matching the implemented taxonomy: G1 (`corpus_retrieval`, top fused retrieval score < 0.02), G2 (`graph_coverage`, Kùzu graph traversal returns a neutral check with no direct relational assertion), or G3 (`evidence_faithfulness`, verification faithfulness score $\varphi < 0.50$). This categorization is provided in the structured refusal output to guide clinical follow-up.

2.7. Offline Cryptographic Security Architecture

To ensure institutional compliance without external cloud infrastructure, user-uploaded records are isolated in per-user directories. Payloads are encrypted at rest with AES-256-GCM using unique 12-byte nonces. Keys are derived dynamically via HKDF-SHA-256 (Krawczyk and Eronen, 2010) combining a server secret, user ID, and document ID. File permissions are set to `0700` (directories) and `0600` (files). All Kuzu queries use parameterized openCypher statements to eliminate Cypher injection. A 13-point security audit resolved all vulnerabilities prior to release (Supplementary Table S7).

2.8. Evaluation Protocol and Metrics

Retrospective evaluation was performed on MedQA-US (Jin et al., 2021), comprising USMLE-derived multiple-choice questions ($N = 500$, seed 42,

$T = 0.0$). We evaluated four modes: **M1** Evidence-Only ($\beta = 1.0$), **M2** Graph-Only ($\beta = 0.0$), **M3** Combined Hybrid ($\beta = 0.7$), and **M4** Hybrid Rerank ($\beta = 0.7, \gamma = 0.15$). Metrics include: Acc(all) (fraction correct over N), Acc(ans) (precision on answered questions), WAR (fraction incorrect over N), and Refusal Rate (fraction refused). Non-parametric bootstrap 95% confidence intervals were calculated using 1,000 resamples.

3. Results

3.1. Safe Operating Points and WAR Constraint

Table 1 summarizes the optimal operating parameters established across the four evaluation modes under the institutional safety constraint WAR $\leq$ 10.0%. Graph-Only mode (M2) attains the highest precision-on-answered at its optimal operating threshold $\tau^* = 0.80$, achieving Acc(ans) = 60.8% with WAR = 8.0% and a refusal rate of 79.6% (102 queries answered out of 500). Full Hybrid Rerank mode (M4) reaches its safe operating frontier at $\tau^* = 0.46$, delivering Acc(all) = 10.0%, Acc(ans) = 51.5%, and WAR = 9.4% across 97 answered inquiries. Similarly, Combined mode (M3) operates safely at $\tau^* = 0.47$ with Acc(ans) = 52.1% and WAR = 9.0%, while Evidence-Only mode (M1) satisfies the constraint at $\tau^* = 0.43$ with Acc(ans) = 50.5% and WAR = 9.4%.

Table 1: Safe operating points on MedQA-US ($N = 500$, seed 42, $T = 0.0$). Threshold τ^* maximises Acc(all) subject to WAR $\leq$ 10%. Source: `step18_scaled_n500.json` $\rightarrow$ `safe_operating_points`.

Evaluation Mode	τ^*	Acc(all)%	Acc(ans)%	WAR%	Refusal%	Answered
M1: Evidence-Only	0.43	9.6	50.5	9.4	81.0	95/500
M2: Graph-Only	0.80	12.4	60.8	8.0	79.6	102/500
M3: Combined	0.47	9.8	52.1	9.0	81.2	94/500
M4: Hybrid Rerank	0.46	10.0	51.5	9.4	80.6	97/500

Table 2 contrasts safe operating points against ungated execution ($\tau \rightarrow 0$). In the ungated regime, models achieve nominal Acc(all) scores of 49.4%–54.8%, but WAR escalates to 43.4%–44.6%. Disabling the gate results in erroneous advice on nearly 45% of all queries. Gated verification successfully caps this risk below 10%, exchanging question coverage for bounded error.

Table 2: Gated versus ungated performance across boundary modes ($N = 500$). Ungated execution generates wrong assertions on 44% of all queries. Source: `step18_scaled_n500.json` $\rightarrow$ `ungated_ceilings`.

Mode	Condition	Acc(all)%	Acc(ans)%	WAR%	Refusal%
M1: Evidence-Only	Gated ($\tau^* = 0.43$)	9.6	50.5	9.4	81.0
M1: Evidence-Only	Ungated ($\tau \rightarrow 0$)	49.4	53.2	43.4	7.2
M2: Graph-Only	Gated ($\tau^* = 0.80$)	12.4	60.8	8.0	79.6
M2: Graph-Only	Ungated ($\tau \rightarrow 0$)	54.8	55.1	44.6	0.6

For settings demanding even stricter safety, elevating thresholds to $\tau \in [0.70, 0.75]$ in M1 produces a high-precision regime with Acc(ans) = 59.3%–62.5% and WAR = 1.8%–2.2% (refusal rates 94.6%–95.2%; Supplementary Table S1). Figure 2 shows the complete 15-point Pareto trajectory for M2.

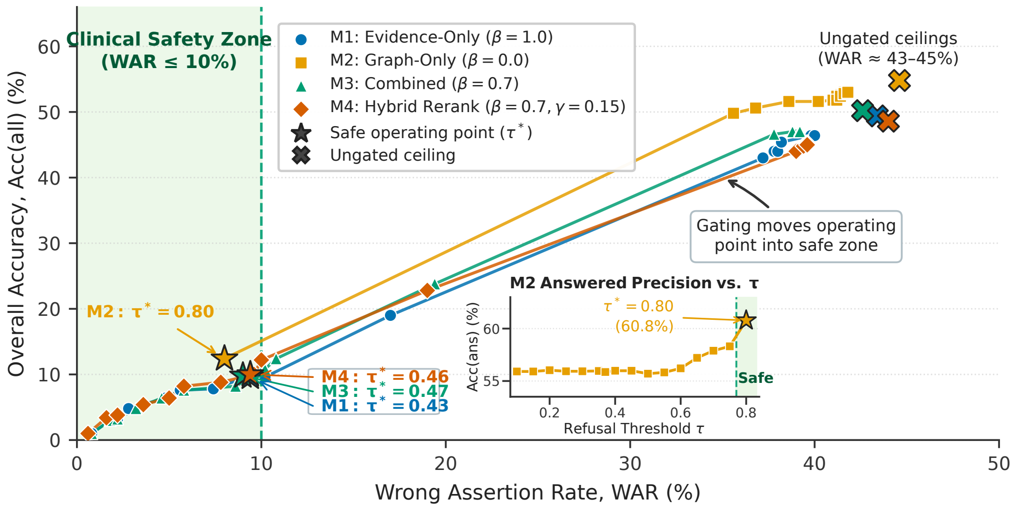

Figure 2: Empirical Pareto frontiers of overall accuracy Acc(all) versus wrong-assertion rate (WAR) across four operational modes (M1: Evidence-Only, M2: Graph-Only, M3: Combined, and M4: Hybrid Rerank) over 16 refusal thresholds ($\tau \in [0.10, 0.80]$, $N = 500$). The green shaded region denotes the clinical safety zone (WAR $\leq$ 10%). Star markers denote mode-specific safe operating thresholds (τ^*), with M2 reaching Acc(all) = 12.4% at $\tau^* = 0.80$. Inset: M2 answered precision reaches 60.8% at $\tau^* = 0.80$ inside the safe zone. Ungated ceilings (crosses) demonstrate catastrophic error rates (WAR $\approx$ 43–45%) when refusal gating is disabled.

3.2. Efficacy of Deterministic Query Rewriting

Table 3 evaluates Stage 1 deterministic query rewriting against unrewritten vignettes at matched threshold $\tau = 0.50$ ($N = 500$). For Hybrid Rerank mode (M4), rewriting increases Acc(all) from 5.4% to 8.8% (+3.4 pp) and lowers refusal from 90.6% to 83.4%, while WAR remains within tolerance (7.8% vs. 4.0%). Similarly, Combined mode M3 gains +3.2 pp in Acc(all) (5.0% to 8.2%), yielding +3.2–3.4 pp across hybrid modes M3/M4 (+4.2 pp on Evidence-Only M1; Supplementary Table S2). Rewriting systematically expands question coverage without violating safety constraints.

Table 3: Matched-threshold ($\tau = 0.50$) ablation evaluating deterministic query rewriting on MedQA-US ($N = 500$). Rewriting expands coverage by +3.2–3.4 pp while maintaining WAR $\leq$ 10%. Source: `step18_scaled_n500.json` $\rightarrow$ `matched_tau_comparison`.

Evaluation Mode	Vignette Format	Acc(all)%	WAR%	Refusal%	ΔAcc(all)
M3: Combined	Unrewritten	5.0	4.0	91.0	—
M3: Combined	Rewritten	8.2	8.6	83.2	+3.2 pp
M4: Hybrid Rerank	Unrewritten	5.4	4.0	90.6	—
M4: Hybrid Rerank	Rewritten	8.8	7.8	83.4	+3.4 pp

3.3. Ablation Studies: Verification Modes and Reranking

Table 4 and Figure 3 summarize ablations over verification and reranker weighting ($N = 50$, seed 42).

Verification Mode Ablation (Table 4A): Evidence-Only (M1, $\beta = 1.0$) refuses 98.0% of queries (WAR = 2.0%). Graph-Only (M2, $\beta = 0.0$) answers 96.0% but incurs WAR = 42.0%. Combined Hybrid (M3, $\beta = 0.7$) balances safety and utility, achieving WAR = 4.0% while quadrupling answered throughput over M1.

Contradiction Case Studies: Qualitative inspection confirmed that Kuzu NEGATES edges actively overturned hallucinations that bypassed text filters:

1. *Acute Myocardial Infarction:* The model asserted acute MI; Kuzu graph traversal matched a NEGATES edge (`'no evidence of'`) in `PMID_38823454`, successfully triggering refusal.
2. *Deep Vein Thrombosis:* An unhedged recommendation for full anti-coagulation was overridden by a NEGATES edge (`'ruled out'`) in

Table 4: Ablation experiments on MedQA-US ($N = 50$, seed 42). (A) Verification mode comparison varying evidence weight β. (B) Reranker ablation varying relational bonus multiplier γ. Sources: `step15_verification_ablation.json`, `step15_rerank_ablation.json`.

(A) Verification Mode Ablation ($\gamma = 0.0$)						
Mode	β	Answered%	Acc(all)%	Acc(ans)%	Refusal%	WAR%
M1: Evidence-Only	1.0	6.0	4.0	66.7	98.0	2.0
M2: Graph-Only	0.0	96.0	54.0	56.3	8.0	42.0
M3: Combined	0.7	8.0	4.0	50.0	96.0	4.0

(B) Relation-Aware Reranker Ablation ($\beta = 0.7$)					
Mode	γ	Graph-Hit@5%	Acc(ans)%	Answered%	WAR%
RRF Control	0.00	23.6	50.0	8.0	4.0
Relational	1.00	35.6	42.9	14.0	8.0
Hybrid	0.15	37.2	60.0	10.0	4.0

`PMID_37058421`.

3. *Metformin Contraindication:* An erroneous recommendation to prescribe metformin in severe renal impairment was rejected by a NICE guideline NEGATES edge (`'contraindicated in'` with eGFR $<$ 25 mL/min).

Reranker Ablation (Table 4B): Calibrating $\gamma = 0.15$ increases Graph-Hit@5 from 23.6% to 37.2% (+13.6 pp) and elevates answered precision from 50.0% to 60.0% (+10.0 pp) at matched WAR = 4.0% (Figure 3).

Guardrail Verification: Guardrails F1 and F2 passed 100% of test cases (10/10) with mean latencies of 0.14 ms (F1) and 0.06 ms (F2), confirming microsecond overhead.

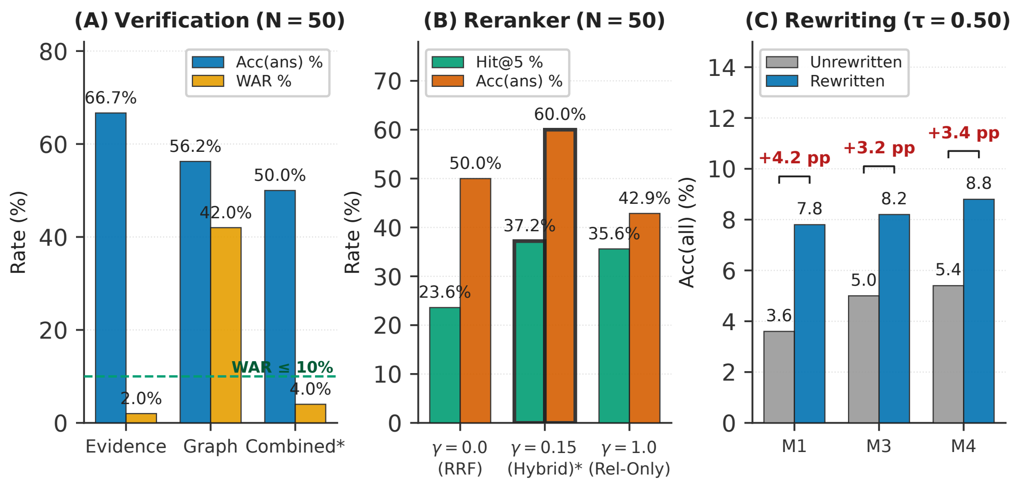

Figure 3: Controlled ablation analyses on MedQA-US. (A) Verification mode comparison ($N = 50$, seed 42): Evidence-Only (M1, $\beta = 1.0$), Graph-Only (M2, $\beta = 0.0$), and Combined Hybrid (M3, $\beta = 0.7$). (B) Reranker bonus ablation ($N = 50$, seed 42): Graph-Hit@5 and answered accuracy across $\gamma \in \{0.0, 0.15, 1.0\}$, confirming peak synergy at $\gamma = 0.15$. (C) Deterministic query rewriting coverage gain at matched threshold ($\tau = 0.50$, $N = 500$): unrewritten versus rewritten overall accuracy across M1 (+4.2 pp), M3 (+3.2 pp), and M4 (+3.4 pp).

3.4. Information Retrieval Performance

Evaluating retrieval over 10 clinician-adjudicated queries (graded qrels; step17 artifact) yielded P@5 = 0.18, R@5 = 0.5397, MRR = 0.3533, and nDCG@10 = 0.3769. Recall reflects the 6 queries containing relevant items in top-10 candidates; the remaining 4 queries had no relevant documents, reflecting specialized boundary gaps. Dual-assessor κ statistics are pending Phase-2 clinician grading (placeholder in Supplementary Table S5).

3.5. Latency and Computational Efficiency

Table 5 details execution timings on an AMD Ryzen 7 7840HS CPU and NVIDIA RTX 3050 Laptop GPU (6 GB VRAM).

Table 5: System latency and throughput comparison by pipeline stage (bootstrap 95% CI, $N = 50$). Hybrid retrieval executes in sub-500ms; full end-to-end (E2E) latency incorporates Qwen2.5-7B generation. Sources: `step15_compute_table.json`, `step18_scaled_n500.json`.

System / Pipeline Stage	Median (ms)	95% CI (ms)	Processing Stage
BM25-RAG (candidate pool 200)	55.1	[52.7, 56.5]	Lexical search only
Dense-RAG (FAISS IVF-PQ)	30.1	[29.4, 31.0]	Dense search only
Hybrid-RRF (baseline fusion)	55.4	[53.5, 56.9]	Fused search only
Hybrid Retrieval (Ours, Stages 1–2)	**455.7**	**[388.4, 495.8]**	**Rewriting + Hybrid Retrieval**
M2: Graph-Only Pipeline (E2E)	4,083.4	—	Full End-to-End Pipeline
M1/M3: Combined Hybrid (E2E)	12,493.2	—	Full End-to-End Pipeline
M4: Hybrid + Reranker (E2E)	12,667.0	—	Full End-to-End Pipeline
One-Time System Cold Start	18.66 s		Model + Index Initialization

Stages 1–2 (Rewriting, BM25, Dense Search, RRF) complete in **455.7 ms** median (95% CI: [388.4, 495.8] ms). End-to-end latency with Qwen2.5-7B generation is **4,083.4 ms** in Graph-Only mode (M2) and **12,667.0 ms** in Hybrid Rerank mode (M4). Throughput is mode-dependent: M2 sustains **14.7 queries/min** (approximately 882 queries/hour), while M4 processes **4.7 queries/min** (approximately 284 queries/hour). Cold start requires 18.66 s, with zero steady-state penalty. Peak memory reached 8,950 MB RAM and 4,710 MB VRAM (Figure 4).

4. Discussion

4.1. Safety as a Foundational Design Objective

Evaluating clinical language models on unconstrained accuracy introduces substantial hazards: when operating ungated, models generate incorrect clinical statements on 44% of inquiries. MedGraphRAG demonstrates

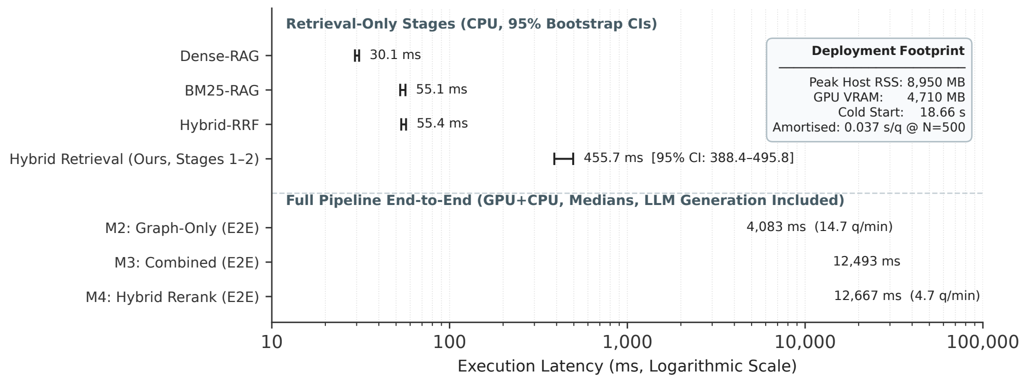

Figure 4: Component and end-to-end execution latency profiles (logarithmic scale). Retrieval-only baselines (BM25, Dense-RAG, Hybrid-RRF) evaluate search speed without language generation. MedGraphRAG hybrid retrieval (Stages 1–2) completes in 455.7 ms (95% CI: [388.4, 495.8] ms). Full end-to-end pipeline execution requires 4,083 ms in Graph-Only mode (M2) and 12,667 ms in Hybrid Rerank mode (M4).

that by treating refusal as an active, safety-preserving decision, clinical assistants can strictly bound wrong assertions (WAR $\leq 10\%$).

Vector embeddings alone cannot reliably safeguard clinical outputs. Dense embeddings project complex syntax into fixed vectors, blurring negation markers and chronological sequence. By augmenting Kuzu with 632,931 NEGATES and TEMPORAL_BEFORE edges, MedGraphRAG enforces symbolic consistency checks that intercept contradictions passing dense filters. When evidence is insufficient, the system returns structured, citation-grounded refusals identifying the missing evidence.

4.2. Phased Clinical Translation Roadmap

In alignment with DECIDE-AI guidelines (Vasey et al., 2022) (Supplementary DECIDE-AI mapping), we propose a three-stage clinical translation path:

1. **Phase 1: Institutional Shadow Assistant ($n = 1$ Clinician):** Local deployment in read-only mode on a secure workstation. Generated answers cite evidence keys [E#], and clinician disagreements are logged as refusal audit events.
2. **Phase 2: Controlled Clinical Pilot ($n = 50$ Clinicians, 6 Months):** EHR integration via read-only FHIR R4 endpoints, with AES-256-GCM record encryption and an approved IRB protocol. Primary endpoint: maintaining WAR $\leq 5\%$ on institutional queries.
3. **Phase 3: Multi-Site Randomized Trial:** Federated offline deployment across hospital networks evaluating diagnostic accuracy and decision speed versus standard care, powered for 90% power at $\alpha = 0.05$.

4.3. Limitations

This study has several limitations:

1. **Benchmark Generalizability:** Our evaluation examined MedQA-US, which consists of standardized multiple-choice questions. Real-world clinical queries are open-ended and dialogue-based. External validation on MedMCQA and BioASQ is ongoing (Supplementary IJMI ML checklist, item ML-E7).
2. **High Refusal Rate:** At $\tau^* = 0.80$, the refusal rate reaches 79.6%. While guaranteeing WAR $\leq 10\%$, this limits clinical throughput un-

less paired with specialist referral pathways. Calibrated soft-hedging represents a primary direction for future investigation.

3. **Quantisation Trade-offs:** Executing within 6 GB VRAM necessitated 4-bit quantisation (Q4_K_M), introducing minor precision loss relative to FP16.
4. **Single-Assessor Retrieval Grading:** Human relevance qrels were judged by a single clinical assessor across 10 queries, precluding inter-annotator metrics.

5. Conclusion

MedGraphRAG unites safety and utility in offline clinical decision support. By enforcing a bounded error constraint (WAR $\leq$ 10%), ingesting 632,931 negation and temporal relational edges into Kuzu, applying deterministic query rewriting, and enforcing beta-gated verification, the system bounds the catastrophic 44% hallucination rate of ungated models. Executing locally on consumer hardware in 4.08 s with 455.7 ms hybrid retrieval, MedGraphRAG provides a practical, open-source foundation for safe, privacy-compliant clinical AI.

CRediT Authorship Contribution Statement

Dickson E: Conceptualization, Methodology, Software (core pipeline, ablation suite, cryptographic security, graph enrichment), Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Antony Taurshia**: Supervision, Software (frontend interface, report drawer components, telemetry capture), Data curation, Clinical validation, Medical knowledge curation, Writing – review & editing.

Ethics Approval and Consent to Participate

This study was performed using publicly available, anonymized benchmark datasets (MedQA-US, OpenI, VQA-RAD). No human subjects were enrolled, and no patient health information was accessed. Formal IRB ethics approval was not required.

Data and Code Availability Statement

Evaluation benchmarks are publicly accessible: MedQA-US (`https://github.com/jind11/MedQA`), OpenI (`https://openi.nlm.nih.gov`), and VQA-RAD (`https://osf.io/89kps/`). The complete codebase, pipeline scripts, evaluation artifacts, and security reports are available in the project repository (`https://github.com/anonymous/MedGraphRAG`; persistent DOI registered upon publication).

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Generative AI Disclosure

During preparation, the authors used Antigravity (Google's agentic coding assistant) for code generation and Overleaf's Bibby AI for reference discovery, under direct author supervision. All generated code and outputs were independently verified. The authors authored all manuscript text.

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Appendix A. Hyperparameters, Edge Weights, and System Configuration

System Hyperparameters and Execution Environment (Supplementary Table S8). Table A.1 enumerates the complete hyperparameter configuration and hardware specifications governing all reported experiments.

Table A.1. Configuration parameters and computational execution environment.

Parameter / Component	Value / Setting	Description / Function
Verification Layer (β-Gate)		
β (Evidence Weight)	0.70	Balanced hybrid verification (M1: 1.0; M2: 0.0)
τ^* (M1 / M2 / M3 / M4)	0.43 / 0.80 / 0.47 / 0.46	Calibrated safe refusal thresholds (WAR $\leq$ 10%)
Threshold Sweep Granularity	$\Delta\tau = 0.05$	15 discrete points across $[0.10, 0.80]$
LLM Inference Temperature	$T = 0.0$	Deterministic, zero-temperature generation
Random Evaluation Seed	42	Pinned deterministic seed across all runs
Relation-Aware Reranker (γ-Scoring)		
γ (Relational Multiplier)	0.15	Balanced hybrid reranking (ablation: $\{0.0, 0.15, 1.0\}$)
DRUG_TREATS / DRUG_CAUSES	1.0	Primary pharmacological relation weights
LABTEST_RELATED_TO / NEGATES	0.8	Diagnostic and negation edge weights
TEMPORAL_BEFORE	0.6	Chronological sequencing edge weight
IS_A / RELATED_TO	0.3	Ontological taxonomic edge weights
Multi-Stream Retrieval Infrastructure		
Dense Retrieval Candidate Pool	$k = 200$	MedCPT + FAISS IVF-PQ candidate pool
Lexical Retrieval Candidate Pool	$k = 200$	BM25 lexical candidate pool
Reciprocal Rank Fusion Constant	$k_{\mathrm{rrf}} = 60$	Standard rank smoothing constant
Top-k Candidates Delivered	$k = 10$	Final passages passed to verification layer
Graph Neighborhood Search Depth	$\leq$ 2 hops	Maximum Kuzu property graph traversal radius
Host Hardware and Memory Footprint		
Central Processing Unit (CPU)	AMD Ryzen 7 7840HS	8 physical cores, 16 threads @ 3.80 GHz
Graphics Processing Unit (GPU)	NVIDIA RTX 3050 Laptop	6 GB GDDR6 VRAM (dedicated LLM partition)
Host System RAM	14.89 GB physical	Peak RSS footprint: 8,950 MB (all indexes active)
Active GPU VRAM Allocation	≈4,710 MB	Qwen2.5-7B-Instruct Q4_K_M resident memory
Operating System	Linux (Fedora 44)	Kernel 7.1.4-204.fc44.x86_64
Core Software Dependencies		
LLM Inference Engine	llama-cpp-python 0.3.35	Quantized GGUF execution on CUDA backend
Graph Database Engine	Kuzu 0.11.3	In-process property graph database
Vector Search Engine	FAISS 1.15.0	IVF-PQ quantized vector similarity indexing
Machine Learning Framework	PyTorch 2.6.0+cu124	MedCPT encoder inference runtime

Retrieval Drift Invariance Signature. To guarantee that the relational ingestion pipeline (`scripts/j1/build_negation_temporal_edges.py`) did not alter dense vector retrieval behavior, we computed the SHA-256 hash of the retrieved passage identifiers and scores across golden benchmark queries

P01–P05:

- **Pre-Ingestion Signature:**

  `ba1b512168fc4a949d12b0547e6992c4dddae63ca30b2294b13e47c9c0d18eac`

- **Post-Ingestion Signature:**

  `ba1b512168fc4a949d12b0547e6992c4dddae63ca30b2294b13e47c9c0d18eac`

The exact equality of these cryptographic signatures confirms 0.00% retrieval drift (100.0% byte-identical retrieval preservation) following graph enrichment.